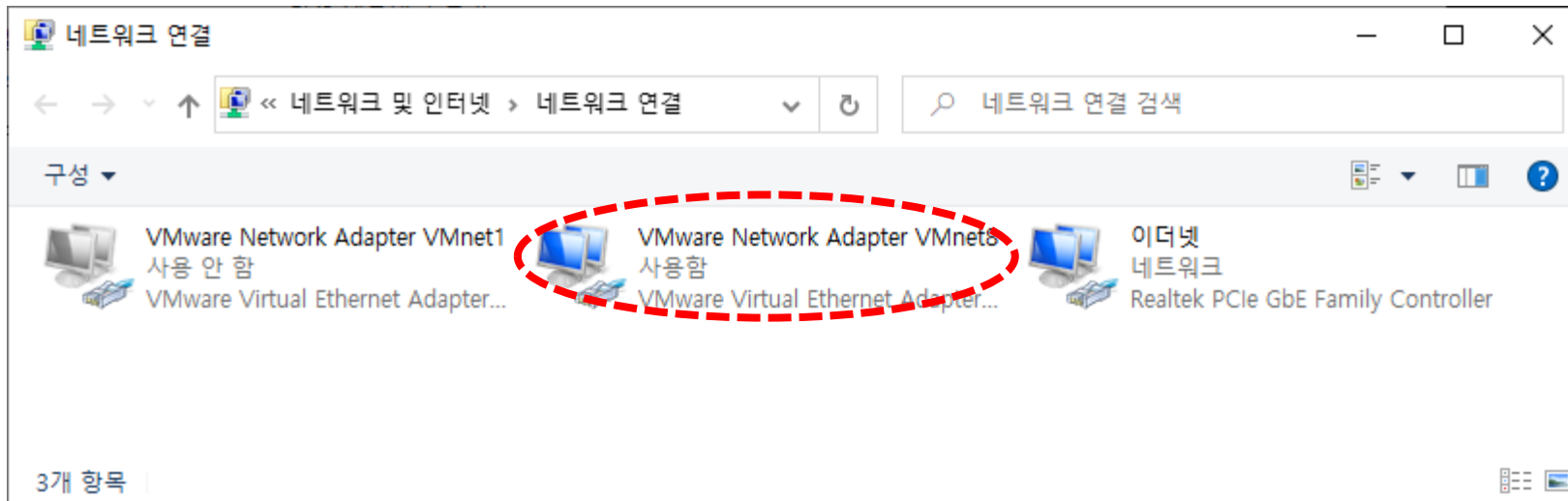
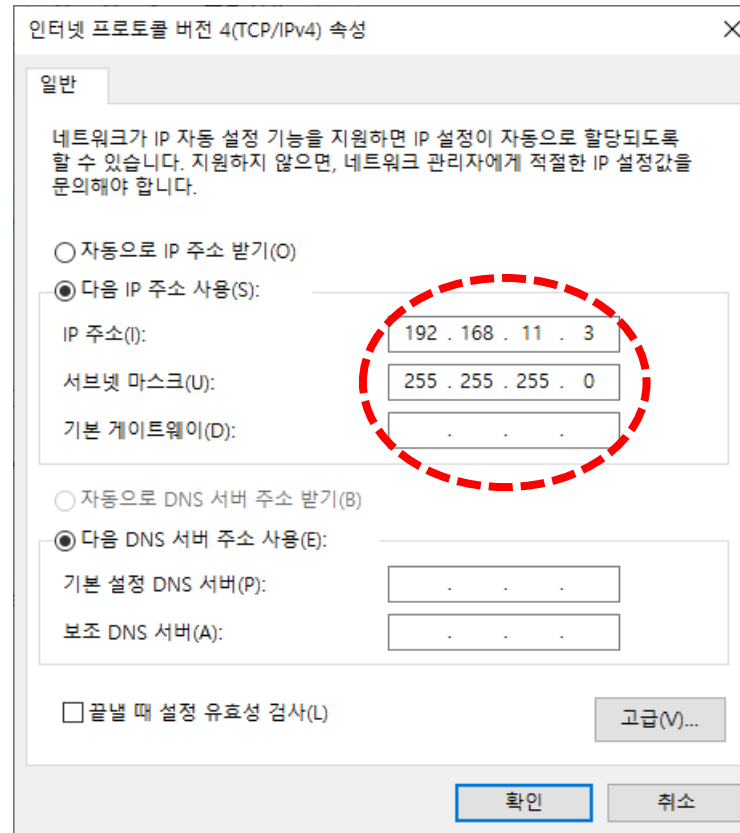
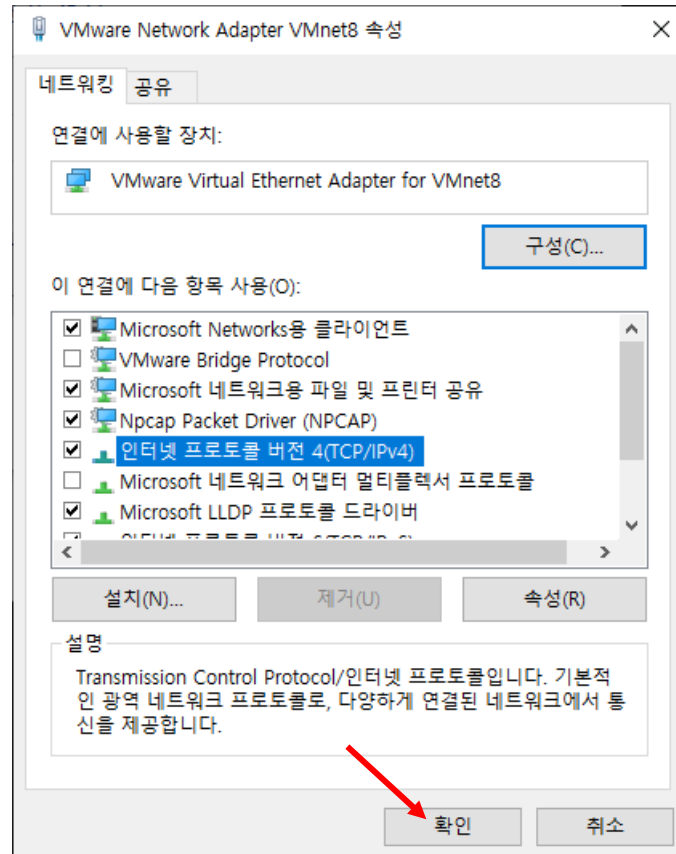

네트워크 NAT 인프라 구성

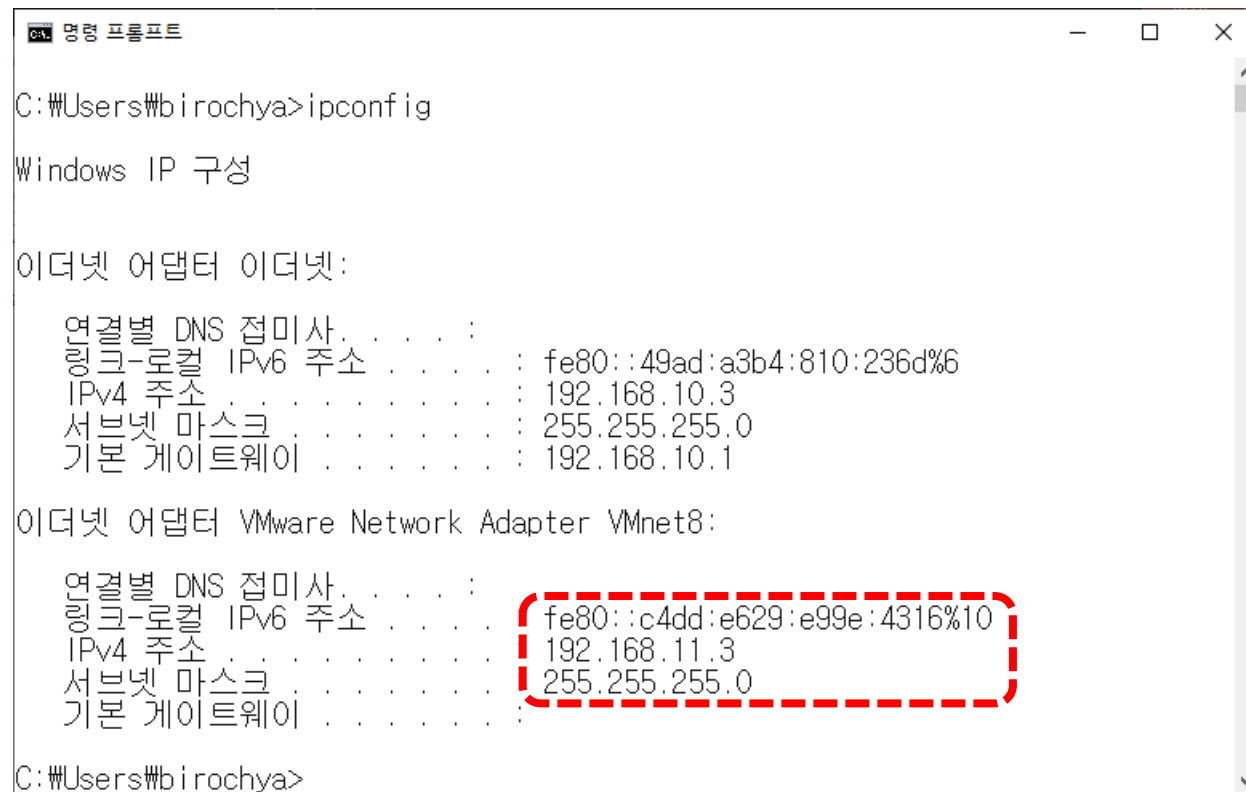
- VMware Network Adapter Vmnet8 설정
[네트워크] ⇒ [속성] → [어댑터 설정 변경]



➤ VMware Network Adapter Vmnet8 설정



➤ VMware Network Adapter Vmnet8 설정



```
C:\Users\birochya>ipconfig

Windows IP 구성

이더넷 어댑터 이더넷:

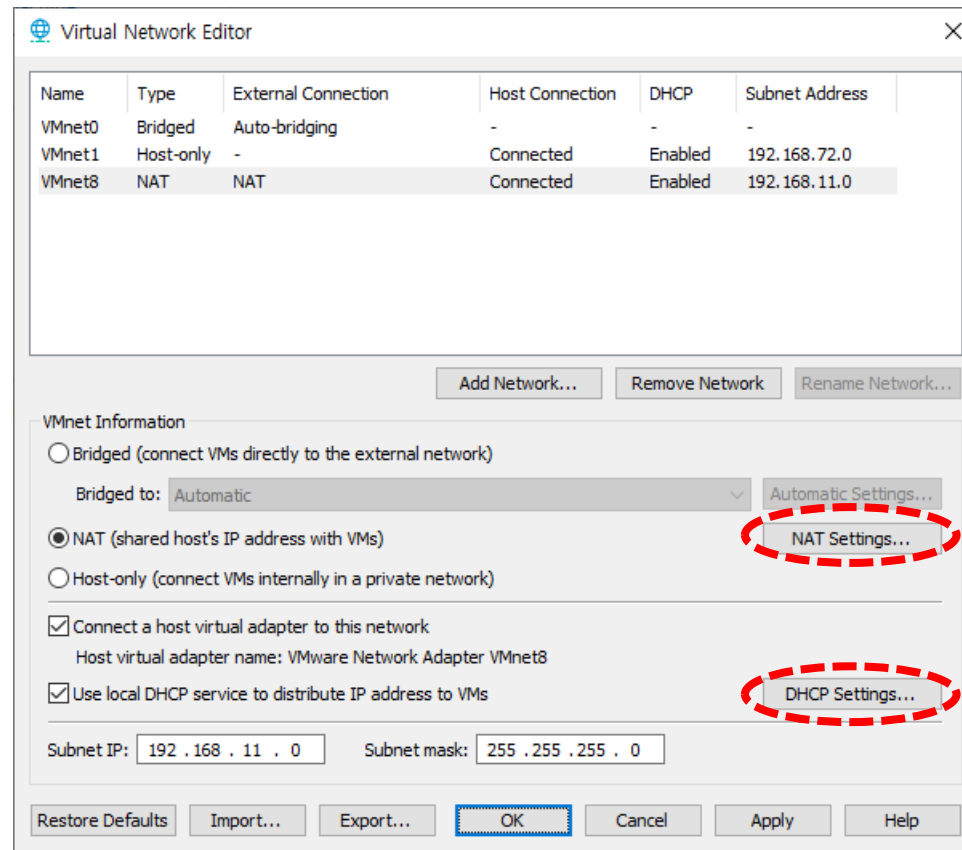
    연결별 DNS 접미사. . . . . : 
    링크-로컬 IPv6 주소 . . . . : fe80::49ad:a3b4:810:236d%6
    IPv4 주소 . . . . . : 192.168.10.3
    서브넷 마스크 . . . . . : 255.255.255.0
    기본 게이트웨이 . . . . . : 192.168.10.1

이더넷 어댑터 VMware Network Adapter VMnet8:

    연결별 DNS 접미사. . . . . : 
    링크-로컬 IPv6 주소 . . . . : fe80::c4dd:e629:e99e:4316%10
    IPv4 주소 . . . . . : 192.168.11.3
    서브넷 마스크 . . . . . : 255.255.255.0
    기본 게이트웨이 . . . . . : 

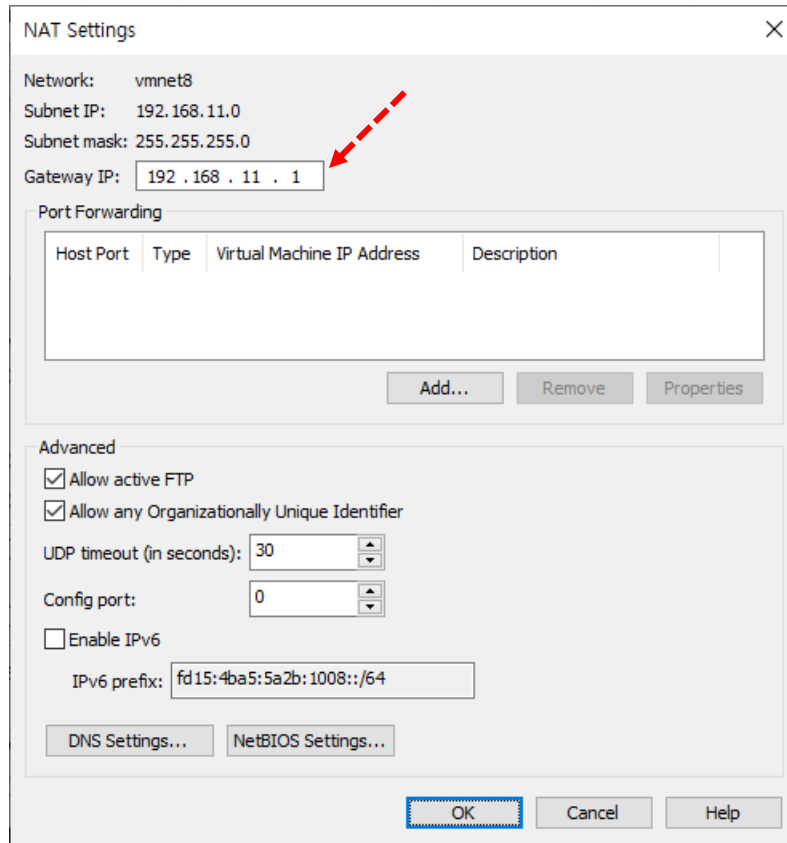
C:\Users\birochya>
```

➤ VMware 설정 : [Edit] → [Virtual Network Editor]



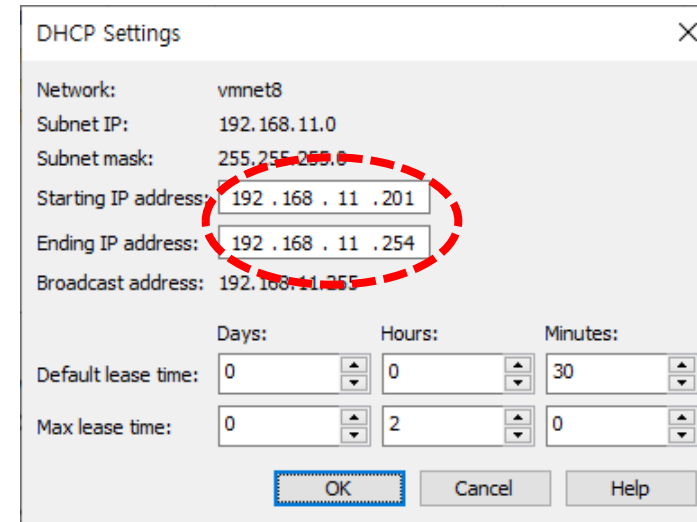
NAT 네트워크 설정

➤ VMware 설정 : [Edit] → [Virtual Network Editor]



The NAT Settings dialog box is shown with the following fields and options:

- Network: vmnet8
- Subnet IP: 192.168.11.0
- Subnet mask: 255.255.255.0
- Gateway IP: 192.168.11.1 (indicated by a red dashed arrow)
- Port Forwarding table with columns: Host Port, Type, Virtual Machine IP Address, Description. Below the table are buttons: Add..., Remove, Properties.
- Advanced section with checkboxes: ☒ Allow active FTP, ☒ Allow any Organizationally Unique Identifier.
- UDP timeout (in seconds): 30
- Config port: 0
- ☐ Enable IPv6
- IPv6 prefix: fd15:4ba5:5a2b:1008::/64
- Buttons at the bottom: DNS Settings..., NetBIOS Settings..., OK, Cancel, Help.



The DHCP Settings dialog box is shown with the following fields and options:

- Network: vmnet8
- Subnet IP: 192.168.11.0
- Subnet mask: 255.255.255.0
- Starting IP address: 192.168.11.201
- Ending IP address: 192.168.11.254 (both the starting and ending IP fields are circled with a red dashed line)
- Broadcast address: 192.168.11.255
- Default lease time: Days: 0, Hours: 0, Minutes: 30
- Max lease time: Days: 0, Hours: 2, Minutes: 0
- Buttons at the bottom: OK, Cancel, Help.

➤ Guest OS 설정 1

▼ Devices	
Memory	1 GB
Processors	1
Hard Disk (SCSI)	40 GB
CD/DVD (IDE)	Auto detect
Network Adapter	NAT
Display	Auto detect

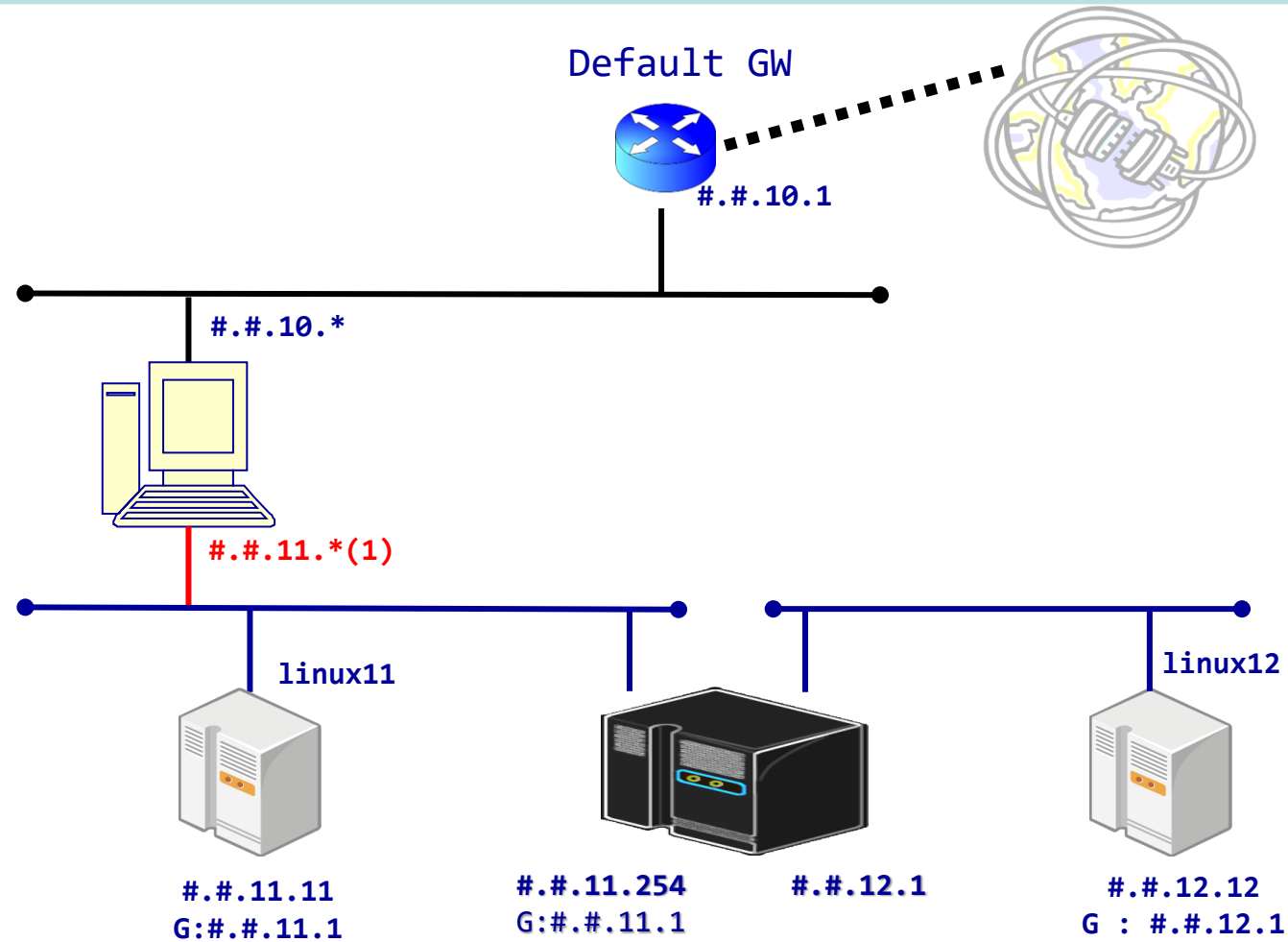
➤ Guest OS 설정 2

IP : 192.168.11.##

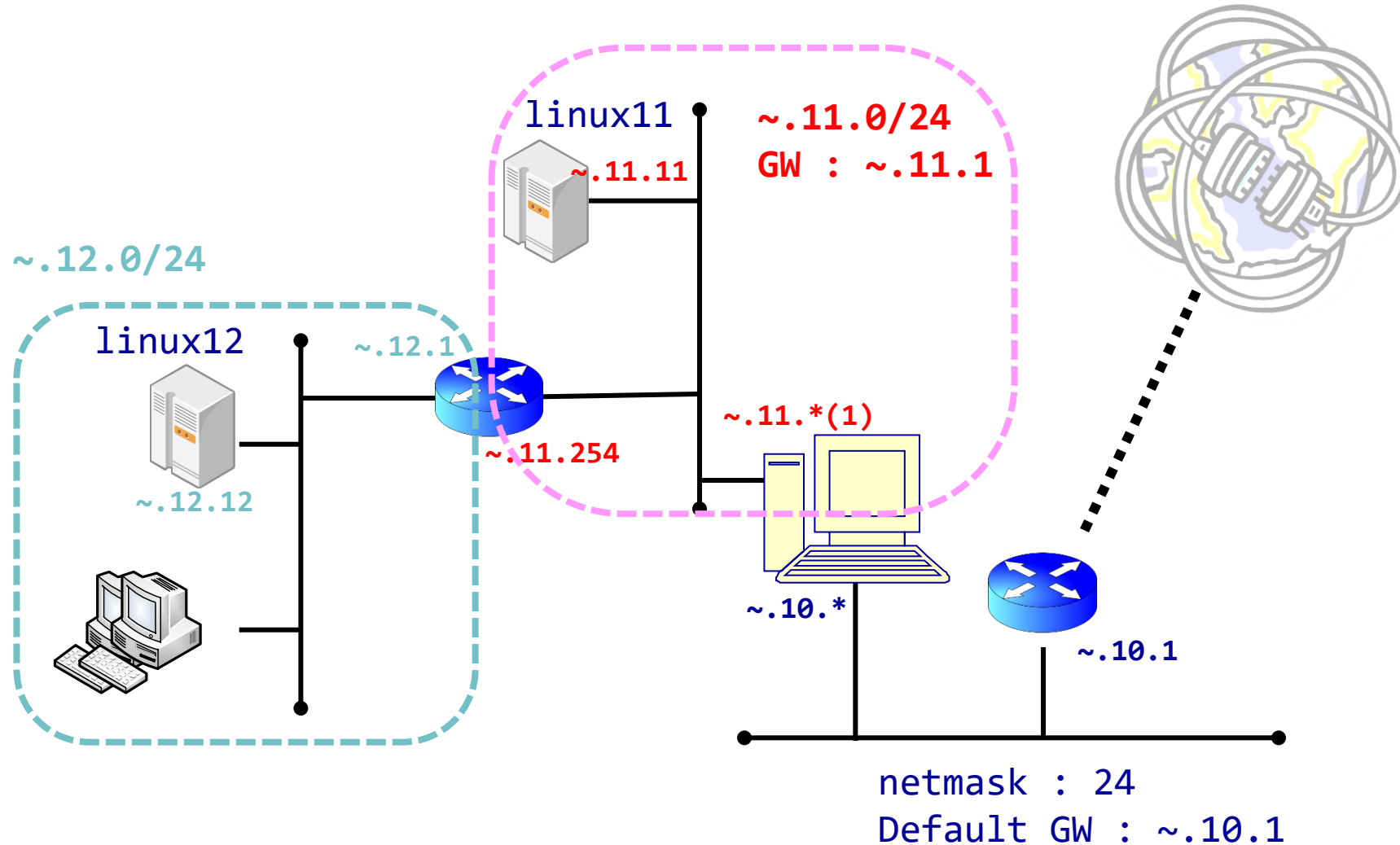
Netmask : 255.255.255.0

Gateway : 192.168.11.1

네트워크 구성 실습 1



네트워크 구성 실습 2 - local router



네트워크 구성 실습 2 - local router

➤ router 리눅스 설정

- 방화벽을 disable 한다.

```
# systemctl disable --now firewalld.service  
# systemctl stat firewalld.service
```

- Selinux를 disable 한다.

/etc/selinux/config 파일의

SELINUX=enforcing → SELINUX=disabled 수정 후 rebooting한다.

```
# sestatus
```

- VMware NAT 실습 시에 NetworkManager stop 하면 리눅스의 네트워크 부분이 작동하지 않을 수 있다.

– NAT 부분에 접속에 문제가 있으면 Vmnet8을 재 시작해본다.

네트워크 구성 실습 2 - local router

- 패킷 포워딩 기능을 활성화 한다.

```
# sysctl -a | grep ipv4.ip_forward
```

```
# sysctl -w net.ipv4.ip_forward=1 (공백없음)
```

or

```
# echo 1 > /proc/sys/net/ipv4/ip_forward
```

or

/etc/sysctl.conf 파일에 net.ipv4.ip_forward = 1

을 수정(추가)한다.

파일 추가 후 sysctl -p 명령을 실행한다.

네트워크 구성 실습 2 - local router

- 10.3 에서 192.168.12.* 네트워크 접속 설정(Windows)

```
route -p add [IP] MASK [mask] [GW_IP]
```

```
route -p add 192.168.12.0 MASK 255.255.255.0 192.168.11.254
```

- 인터페이스가 잘못 지정되는 경우 IF를 설정한다.

- 11.## 에서 192.168.12.* 네트워크 접속 설정(Linux)

```
ip route add 192.168.12.0/24 via 192.168.11.254 dev ens160
```